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When are artificial intelligence versus human agents faulted for wrongdoing? Moral attributions after individual and joint decisions

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ABSTRACT

Artificial intelligence (AI) agents make decisions that affect individuals and society which can produce outcomes traditionally considered moral violations if performed by humans. Do people attribute the same moral permissibility and fault to AIs and humans when each produces the same moral violation outcome? Additionally, how do people attribute morality when the AI and human are jointly making the decision which produces that violation? We investigate these questions with an experiment that manipulates written descriptions of four real-world scenarios where, originally, a violation outcome was produced by an AI. Our decision-making structures include individual decision-making – either AIs or humans – and joint decision-making – either humans monitoring AIs or AIs recommending to humans. We find that the decision-making structure has little effect on morally faulting AIs, but that humans who monitor AIs are faulted less than solo humans and humans receiving recommendations. Furthermore, people attribute more permission and less fault to AIs compared to humans for the violation in both joint decision-making structures. The blame for joint AI-human wrongdoing suggests the potential for strategic scapegoating of AIs for human moral failings and the need for future research on AI-human teams.

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Introduction

Humans working together can accomplish amazing things that would be impossible for an individual, making human teams invaluable in modern society. Teams, however, possess just as great of a potential for morally unacceptable behavior. A long history of social science research shows how individuals are attributed responsibility, blame, praise, or wrongness (Malle, Guglielmo, & Monroe, 2014; Shaver, 1985; Weiner, 1985) and this has also been applied to humans in teams and organizations (Gailey, 2013; Haynes & Lawrence, 2012; Shank & Robinson, *forthcoming*). An emerging question is whether similar attributions can be made to non-humans such as intelligent technologies which now routinely collaborate with humans in joint decision-making, thereby functioning as rudimentary teams.

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Assigning moral fault within a group of humans may involve more than a logical calculus of determining responsibility, but may depend on relative power and status (Shank & Robinson, [forthcoming](#)), group identification (Kaiser, Dyrenforth, & Hagiwara, 2006), and motives for scapegoating (Rothschild, Landau, Sullivan, & Keefer, 2012). Because artificial intelligence (AI) agents will often have little social power and may be thought of as tools instead of team members, if people do blame AIs on human-AI teams there may be increased scapegoating of moral responsibility. This could have a profound implication for our society as AIs continue to join us as teammates at work, while simultaneously being feared for taking our jobs (McClure, 2018).

Specific situations where AIs are involved in morally significant outcomes that have gained media attention include AIs producing lenient parole sentences towards whites, selecting disproportionately few dark-skinned winners in a beauty contest, exposing children to violent and obscene videos, and producing derogatory remarks about minority groups on Twitter (Shank & DeSanti, 2018). More generally, there is widespread use in society of black-box algorithms which routinely discriminate against minority groups and the poor (Eubanks, 2018; Noble, 2018; Wachter-Boettcher, 2017) and are used to justify their own performance and cutthroat business practices in the name of efficiency (Eubanks, 2018; O'Neil, 2016). Some evidence shows that humans attribute some moral fault to the AIs (Shank & DeSanti, 2018), although perhaps at a reduced level compared to humans (Malle, Magar, & Scheutz, [forthcoming](#); Pizarro, 2014). However, there have been few direct comparisons of moral fault attributed to AIs versus humans engaged in the exact same real-world moral violations (e.g., Malle et al., [forthcoming compares humans and machines in a military strike dilemma](#)). Therefore, the first purpose of our research is to *determine the attribution of moral permissibility and fault of AIs versus humans in real-world moral violation situations*.

Humans team up with artificial intelligence to jointly make decisions in several different structural arrangements. In situations where an AI functions with relative autonomy, but is monitored by a human, such as self-driving cars, it is unclear who would be to blame if people are injured in an accident (McManus & Rutchick, 2018). This problem is compounded when the car can decide to change course and injure one person in a crosswalk to avoid injuring many on the side of the road (Bonnefon, Shariff, & Rahwan, 2016; Nyholm & Smids, 2016). Aside from autonomous vehicles, AIs make autonomous decisions monitored by humans on Twitter, in production lines, in stock trading, in automatic bids, and with military drones. In other situations, an AI may make recommendations, such as providing doctors with the most probable diagnoses for patients. Would the doctor be at fault if the AI were wrong? While expert systems have been able to recommend probable diagnoses of disease for over four decades (Shortliffe et al., 1975), the US FDA only recently approved these AIs to be marketed as medical devices (US Food & Drug Administration, 2018). Aside from medical recommendations, AIs recommend books, movies, people to date, applicants to hire, clients to give loans to, and products to buy (Willson, 2017). Therefore, studying moral attributions to humans and AIs in the form of monitoring and recommending structures mirrors two of the most common joint decision-making arrangements in society.

When humans make decisions resulting in a moral violation, their severity of blame is determined primarily by the degree of autonomy with which their decision was made (Gailey & Falk, 2008; Malle et al., 2014; Malle & Knobe, 1997). If this idea carries over

to human-AI joint decision-making, it would suggest that the AI being monitored would be at greater fault than the human doing the monitoring, and that the human receiving recommendations would be at greater fault than the AI making the recommendations. Therefore, the second purpose of our research is to *investigate the attribution of moral fault and permissibility in joint-decision making structures of an AI monitored by a human and of a human receiving recommendations from an AI.*

Distinguishing levels of moral judgement

In situations with AIs acting, it is important to consider different levels at which moral judgements may occur. At the level of the situation, there is the question of general wrongness – whether or not any moral wrong occurred. Before one can make attributions to a human, AI, or other causal agent, there must first be some judgement or outcome necessitating a cause (Kelley, 1972; Malle et al., 2014; Weiner, 1986). It is possible that outcomes which are generally agreed to be morally wrong if caused by a human will not be seen as wrong if caused by an AI. Yet, in situations where moral violations are caused by AIs, many people make judgements of general wrongness or fault (Malle et al., *forthcoming*; Shank & DeSanti, 2018) which also reflect a violation of the expected moral foundation (Shank & DeSanti, 2018) – care, fairness, loyalty, authority, purity, and liberty (Graham et al., 2011). However, it is possible those judgements and the moral foundation violated differ from those made to humans engaged in the same behavior.

At the level of specific attributions to specific entities, there is a difference between attributions of moral permissibility and moral fault. Permissibility attributions typically occur before an action and are focused on evaluating options to act, whereas wrongness and blame are part of the evaluating fault for a violation after it occurs (Malle et al., 2014). Among both humans and robots, some evidence points to permissibility attributions being based on the consequences of the action, blame attributions based on mental agency factors, and wrongness attributions based on both (Voiklis, Kim, Cusimano, & Malle, 2016). Other evidence shows that judgements of general wrongness when AIs are the proximate actors in a violation outcome are negatively correlated to perceiving the AI as justified (i.e., that it is a permissible behavior) and that the AI is wrong (i.e., that the AI is at fault; Shank & DeSanti, 2018). Even in cases where people refuse to fault the AI due to its lack of capacity to produce the desired outcome, they may still find that it not morally permissible (Malle et al., 2014; Voiklis et al., 2016). Therefore, we primarily focus on fault and permissibility, but we also consider that other moral attributions such as awareness, responsibility, and intentionality which are distinct from fault and permissibility under many circumstances (Gailey & Falk, 2008; Malle et al., 2014; Malle & Knobe, 1997; Shaver, 1985).

Moral attributions to AIs versus humans

Moral attributions are readily made to AIs when they are the proximate cause of an outcome that would normatively be considered a moral violation if perpetrated by a human (Malle, Scheutz, Arnold, Voiklis, & Cusimano, 2015; Shank & DeSanti, 2018). Furthermore, mirroring findings about interactions with computer agents in general (Reeves & Nass, 1996), neither the organization that owns or uses the AI, the developers who programmed it, nor the users that interact with it are attributed moral fault for a moral

violation (Shank & DeSanti, 2018). However, there is sufficient evidence to suggest that AIs may generally be attributed *less* fault than humans. Artificial agents are blamed less and attributed less wrongness than humans for making the same moral choices in a trolley-problem ethical dilemma (Voiklis et al., 2016). Similarly, attributions of goodness and power are attenuated for computer agents compared to human agents (Shank, 2013, 2014). In contrast, machines are sometimes blamed more than humans, such as when artificial teammates make mistakes (Merritt et al., 2011) or when machines fail to take action in a trolley-problem (Malle et al., 2015). These cases of high AI blame seem to primarily be due to specific expectations about the AIs acting in an efficient, optimal, and rational way, at least compared to humans.

Moral attributions in joint decision-making structures

In joint decisions, the nature of the agents, human versus AI, could be definitive in moral attribution. However, the structural arrangement of the decision making may also influence attributions in three ways. First, the agent that is most proximate to the decision may be faulted more than less proximate agents (Lawler, 1992; Lawler, Thye, & Yoon, 2016; Shank & Robinson, *forthcoming*). Second, in attribution theory terms the most proximate agent would typically be evaluated in terms of the locus of control, and only if the cause appeared external to that agent would the attribution be shifted (Weiner, 1985). Third, moral faults may be more associated with acts of commission versus omission. If an agent caused a violation outcome through their inaction or omission, the fault assigned to them will generally be less than if they caused it through their action (Malle et al., 2015; Voiklis et al., 2016). Therefore, in cases where multiple entities are making a joint decision, the one most directly associated with committing the action should receive greater moral scrutiny. All three arguments point to AIs in the monitoring structure and humans in the recommendation structure being more at fault than their counterpart due to their proximate position as the one generating and submitting the decision and thereby committing the moral fault.

Experiment

Amazon Mechanical Turk (mturk) participants completed an online experiment where they were randomly assigned to one of 32 conditions formed by a full factorial crossing of 4 events with 4 decision-making structures and with 2 outcomes (control or violation). The four decision-making structures varied the presence and role of humans and AIs with decisions made by (1) a solo AI, (2) a solo human, (3) an AI with a human monitoring it (monitoring), and (4) a human provided recommendations by an AI (recommending). The control outcomes are used to show how the violation outcomes differ in general wrongness, whereas the violation outcomes are solely used to investigate attributions of those violations.

Event, structure, and outcome conditions

We presented events that were based on four actual events where an artificial intelligence or advanced algorithm produced an outcome which would traditionally constitute a moral violation. These events were previously adapted from media coverage into neutral

language vignettes that could be used as experimental stimuli (Shank & DeSanti, 2018). We selected four of Shank and DeSanti's vignettes that made sense for humans as well as AIs and we adapted them to create the human, monitoring, and recommending conditions. The vignettes also originally included information about the organization which controlled the AI software. Shank and DeSanti (2018) found that respondents' familiarity with the organization altered their perceptions of moral fault, yet neither the organization, the software developers, nor the users were blamed for the wrong. Therefore, we simplified the vignettes by removing information about the organization which controlled the AI.

The *parole sentencing* event describes predicting recidivism rates of convicts in order to determine a parole sentencing. In the violation condition the sentence is unfairly lenient for whites and harsh for blacks. The *beauty contest* event involves determining the winners of a beauty pageant. In the violation condition the winners are primarily white and light skinned. The *children's videos* event involves a video-playing mobile app developed specifically for children. In the violation condition the videos included obscene and violent material. The *Twitter* event involves tweets from one account. In the violation condition the tweets contained hate speech. The control conditions for all events were the expected outcome of such an event (see online supplemental data for exact wording). For both control and violation outcomes, the wording identified what specifically the outcome was, so that the moral attribution questions would be focused specifically on the outcome. The 32 vignettes ranged from 34–109 words long and a reading grade level from 6.6–11.8.

Participants and procedure

The 453 participants of this study were recruited through a general posting on mturk. Amazon Mechanical Turk is an online crowdsourcing micromarket where people can post tasks (called human interaction tasks) that others will claim and complete for a small payment. The uses of crowdsourcing platforms and crowdsourcing markets are varied (Shepherd, 2012), but increasingly academics use them to recruit participants for academic research surveys and experiments (Buhrmester, Kwang, & Gosling, 2011; Mason & Suri, 2012) as well as other types of academic research (Shank, 2016). While they are not perfect samples, the sample is often more similar to the US population than studies using university students (Buhrmester et al., 2011; Mason & Suri, 2012). In order to minimize participants who were inattentive, participants were required to have a positive history on mturk, meaning 90%+ of their work accepted and at least 100 tasks completed (Mason & Suri, 2012; Shank, 2016). The survey was completed in Qualtrics, took an average of 5–6 min to complete, and had a compensation of \$0.60. Based on random assignment to a condition participants were shown one event each. This was followed by questions concerning moral attributions, moral foundations, familiarity with the event, and general demographic information.

Measures

Moral violation attributions

Seven items measured moral attributions (Gailey, 2013; Malle et al., 2014), with a four-point scale each with specific labels corresponding to none (1), somewhat (2), moderately (3), and completely (4) attributing that variable to the agent. For the monitoring and

recommending conditions both the human and AI were listed and responded to separately, whereas the human and AI condition each had only one respective agent.

- (1) Responsibility: How responsible is the following for the outcome?
- (2) Awareness: Was the following aware of the potential for this outcome?
- (3) Intention: Did the following intend for this outcome to occur?
- (4) Justified: Was the following justified in their role in the outcome? (reversed)
- (5) Permissible: How permissible is it for the following to produce this outcome? (reversed)
- (6) Blame: How much do you blame the following for the outcome?
- (7) Wrongness: How wrong was the following for their role in the outcome?

We subjected the seven moral attributions items to a factor analyses, considering components with Eigenvalues ≥ 1 and items with loadings of over 0.5. For attributions of the violation outcome to both humans and AIs two components emerged: *moral fault* is the average of wrongness, blame, responsibility, awareness, intentionality ($\alpha = .76$ for human and $\alpha = .78$ for AIs) whereas *moral permissibility* is the average of permissible and justified ($\alpha = .77$ for human and $\alpha = .74$ for AIs). The attributions of the normatively expected control outcomes failed to produce consistent and interpretable scales via factor analyses. Many of these moral attributions to individual agents lack coherent meaning in the absence of a violation outcome and therefore the control condition is not used for analysis of specific attributions (Gailey & Falk, 2008; Malle et al., 2014; Shank & DeSanti, 2018).

Other measures

Participants were also asked the degree of moral wrongness occurring in the event by responding to a five-point scale ranging from *not at all* (1) to *very much* (5) in response to *For this event, how much did each of the following occur?* General wrongness was in response to *A moral violation or wrong*, and the six moral foundations were in response to *harm or causing pain*, *unfairness or cheating*, *disloyalty or betrayal*, *disrespecting or disobeying authority*, *degrading or impure actions*, and *restriction of liberty or loss of privacy*. All seven were presented in a random order.

Participants were asked *Have you heard about this scenario before reading it here?* and answered with yes, unsure, or no. Few people were familiar with the events and preliminary analyses indicated including familiarity did not substantively change the results. Therefore, familiarity was not reported in the results. Demographic questions were the final survey items.

Results

The sample of 453 participants (207 males, 244 women, 2 no response) ranged in age from 18 to 75 (mean of 36.2), were primarily white (73.4% Caucasian/white; 10.2% African American/black; 8.6% Asian/Pacific Islander; 5.5% Hispanic; 2.4% other/no response), and were predominately college educated (10.4% no college; 38.4% some college; 51.0% at least bachelor's degree; 2 did not respond). Respondents were randomly assigned to one of the 32 conditions resulting in 13–16 participants in each. No data was excluded.

Moral perceptions of the situation

How do people perceive the general wrongness across these situations?

Before examining moral attributions to humans and AIs, we examined the moral wrongness across experimental conditions. We conducted a full-factorial MANOVA predicting whether the event was generally perceived as morally wrong and whether it violated each moral foundation (Table 1).

First, as expected we found that the violation outcomes produce more general wrongness (3.48; Table 2) than the control (2.30), $F(1,415) = 96.43$, $p \leq .001$, but that the events also differ in general wrongness, $F(3,415) = 7.89$, $p \leq .001$. Examining these in more detail, we split the control and violation outcome conditions and conducted a MANOVA predicting event general wrongness (and moral foundation violations) by event including post hoc pairwise comparisons between each pair of events. These indicated that general wrongness did not vary by event in the control condition, $F(3,222) \leq 2.03$, ns (see Table 2 for significant pairwise comparisons), but did differ by event in the violation condition, $F(3,217) = 6.11$, $p \leq .001$. This was due to a violation in the beauty contest event being perceived as less wrong (2.88) than the other events (3.59–3.84; Table 2). There were no other significant effects on general event wrongness (Table 1), therefore there is no evidence that the presence or structure of AIs and humans influences the general perceptions of the commission of a moral wrong.

How do people perceive moral foundation violations across these situations?

The violation outcome produced greater violations for five of the six moral foundations, $F_s(1,415) \geq 18.97$, $ps \leq .001$; Table 1, and marginally more oppression, $F(1,415) = 3.21$, $p = .074$. The specific events also differed in all six moral foundation violations, $F_s(3,415) \geq 3.01$, $ps \leq .03$, and an interaction effect between event and violation outcome influenced unfairness, degradation, and oppression $F_s(3,415) \geq 4.80$, $ps \leq .003$; Table 1. Divided by violation and control outcomes, MANOVAs indicate that while there are differences in the control outcome conditions on betrayal, $F(3,222) = 2.695$, $p = .047$, and oppression, $F(3,222) = 5.026$, $p = .002$, there are differences in the violation outcome conditions for all six moral foundations, $F_s(3,217) \geq 3.406$, $ps \leq .019$; Table 2.

The specific differences correspond to our face-valid expectations and prior research results (Shank & DeSanti, 2018): the parole sentencing violation is perceived as highly unfair, and oppressive, the beauty contest violation as unfair, and the children's videos and Twitter violations as degrading (Table 2). Mirroring the general wrongness result, the beauty contest violation had significantly lower means than most of the other events

Table 1. F-statistics from a MANOVA of the general moral wrong and specific moral foundation wrongs by experimental conditions.

	Wrong	Harm	Unfair	Betrayal	Subversion	Degradation	Oppression
Event	7.89***	3.01*	11.78***	5.58***	6.04***	5.99***	24.09***
Violation	96.43***	59.95***	31.64***	18.97***	20.02***	83.01***	3.21†
Decision-Making Structure	1.00	.74	.60	.83	1.43	1.10	.03
Event * Violation	.60	1.02	8.27***	1.79	1.61	4.80**	5.16**
Event * Structure	.98	1.10	.80	1.63	1.48	.64	.90
Violation * Structure	.40	.14	.07	.53	.18	.28	.86
Event * Violation * Structure	1.07	1.06	1.06	1.31	1.29	.36	1.60

Note: $N = 447$. *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$.

Table 2. Means (standard deviations) by experimental conditions.

	Total	F-statistic	Events			
			Parole Sentencing	Beauty Contest	Children's Videos	Twitter
Control ($N = 226$)						
General Wrongness	2.30 (1.26)	2.03	2.38 ^{ab} (1.17)	1.96 ^a (1.16)	2.50 ^b (1.40)	2.34 ^{ab} (1.27)
Harm	2.18 (1.23)	3.10	2.21 (1.15)	2.07 (1.23)	2.16 (1.26)	2.27 (1.31)
Unfair	1.98 (1.13)	.81	2.07 (1.06)	1.96 (1.19)	1.79 (1.07)	2.10 (1.21)
Betrayal	1.87 (1.16)	2.70*	1.77 ^{ab} (1.13)	1.57 ^a (0.93)	2.02 ^b (1.27)	2.10 ^b (1.23)
Subversion	1.88 (1.09)	1.03	1.95 (1.11)	1.66 (1.01)	1.93 (1.12)	1.98 (1.11)
Degradation	2.11 (1.22)	1.30	1.84 (1.11)	2.21 (1.25)	2.20 (1.29)	2.20 (1.24)
Oppression	2.32 (1.33)	5.03**	2.74 ^a (1.19)	1.91 ^b (1.20)	2.07 ^{bc} (1.33)	2.54 ^{ac} (1.45)
Violation ($N = 221$)						
General Wrongness	3.48 (1.30)	6.11***	3.60 ^a (1.15)	2.88 ^b (1.49)	3.84 ^a (1.11)	3.59 ^a (1.23)
Harm	3.12 (1.31)	3.41*	3.38 ^a (1.32)	2.68 ^b (1.44)	3.09 ^{ab} (1.20)	3.34 ^a (1.20)
Unfair	2.62 (1.39)	17.19***	3.56 ^a (1.26)	2.82 ^b (1.40)	1.95 ^c (1.20)	2.24 ^c (1.16)
Betrayal	2.42 (1.40)	3.81*	2.74 ^a (1.44)	1.91 ^b (1.30)	2.66 ^a (1.40)	2.37 ^{ab} (1.35)
Subversion	2.42 (1.43)	5.23**	2.74 ^a (1.47)	1.84 ^b (1.26)	2.34 ^{ab} (1.43)	2.78 ^a (1.38)
Degradation	3.25 (1.37)	9.12***	3.10 ^a (1.40)	2.55 ^b (1.45)	3.66 ^c (1.15)	3.64 ^c (1.20)
Oppression	2.53 (1.43)	24.48***	3.75 ^a (1.12)	2.09 ^{bc} (1.38)	1.86 ^b (1.09)	2.47 ^c (1.33)

Note: F-statistics is difference among for events per row: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$. Event means per row that have any superscript in common or no superscripts at all do not significantly differ according to a post hoc pairwise comparison using the least squared difference.

for harm, betrayal, subversion, and degradation (Table 2). We conclude that our selected events were perceived as generally wrong and specifically violated the expected moral foundation after a violation outcome, but not after the control outcome. Similar to general wrongness, there were no differences in moral foundation violations based on human or AI presence or structure. This bolsters our confidence that any moral attribution differences based on the presence of AIs and humans and their decision-making structure is not explainable by people differently interpreting the morality of the entire situation.

Moral attributions of the violation outcomes to humans and AIs

Next, we consider the moral permissibility and moral fault attributed to human and AI agents in the different decision-making structures after a violation outcome. Because events differ in general levels of moral wrongness as indicated above, we continue to use events as factors in our analyses to reveal any attribution-level difference among events.

How do people attribute moral violations to AIs versus humans?

We compared the moral violation attributions toward the AI in the solo AI conditions to the human in the solo human conditions (Table 3 and Figure 1) using an agent (human or AI) by event MANOVA (Table 4). There are no significant differences in the moral permissibility between humans and AI agents $F(1,108) = 1.79$, ns, and among events, $F(3,108) = 1.36$, ns. However, between violation-producing agents, humans are morally faulted (3.07) more than AIs (2.00), $F(1,108) = 63.82$, $p \leq .001$. While events do not independently alter moral fault, the agent difference is qualified by an interaction effect with events, $F(3,108) = 6.20$, $p \leq .001$. Humans are always faulted more than AIs, but the magnitude of this difference is situationally dependent based on the event: children's videos (.39), parole sentencing (.67), beauty contest (1.33), and Twitter (1.83). Additionally,

Table 3. Means (standard deviations) of moral attributions to AIs by experimental conditions.

Decision-Making Structure	Agent	N	Parole Sentencing	Beauty Contest	Children's Videos	Twitter	All Events
Moral Permissible							
Solo AI	AI	58	1.92 (0.98)	2.39 (1.11)	2.03 (1.06)	2.25 (1.06)	2.16 (1.04)
Solo Human	Human	58	1.75 (0.78)	2.13 (1.23)	1.64 (0.69)	2.07 (1.03)	1.91 (0.96)
Monitoring	AI	55	2.23 (0.83)	2.61 (1.02)	2.18 (1.19)	2.32 (1.17)	2.34 (1.05)
Monitoring	Human	55	2.12 (1.00)	2.29 (0.97)	1.86 (1.08)	1.75 (0.94)	2.00 (1.00)
Recommending	AI	53	2.58 (0.81)	3.04 (0.97)	1.73 (0.67)	1.82 (1.07)	2.28 (1.03)
Recommending	Human	53	2.62 (0.82)	1.62 (0.89)	1.46 (0.69)	1.54 (0.60)	1.80 (0.87)
All Structures	AI	166	2.24 (0.89)	2.67 (1.05)	1.99 (1.00)	2.14 (1.10)	2.26 (1.04)
All Structures	Human	166	2.15 (0.92)	2.02 (1.06)	1.66 (0.84)	1.79 (0.89)	1.90 (0.94)
Moral Fault							
Solo AI	AI	58	2.00 (0.73)	1.81 (0.53)	2.51 (0.70)	1.69 (0.79)	2.00 (0.75)
Solo Human	Human	58	2.67 (0.70)	3.15 (0.64)	2.90 (0.61)	3.52 (0.89)	3.07 (0.77)
Monitoring	AI	55	2.05 (0.76)	2.11 (0.55)	2.03 (0.75)	1.63 (0.77)	1.95 (0.72)
Monitoring	Human	55	2.85 (0.78)	2.41 (0.56)	3.16 (0.56)	2.69 (0.80)	2.77 (0.72)
Recommending	AI	53	2.26 (0.90)	1.55 (0.41)	2.37 (0.72)	1.69 (0.71)	1.96 (0.77)
Recommending	Human	53	3.17 (0.52)	2.85 (0.82)	3.22 (0.40)	3.07 (0.55)	3.08 (0.59)
All Structures	AI	166	2.10 (0.79)	1.83 (0.54)	2.30 (0.74)	1.67 (0.74)	1.97 (0.74)
All Structures	Human	166	2.89 (0.69)	2.81 (0.73)	3.09 (0.54)	3.10 (0.82)	2.97 (0.71)
N			40	42	42	44	166

Note: Violation conditions only ($N = 221$).

the latter two with the largest difference appear to be caused by both more fault attributed to humans *and* less fault attributed to AIs compared to the other conditions (Figure 2). We return to this interesting result in the discussion.

How do people attribute moral violations to AIs in different decision-making structures?

We compared the moral violations attributed to AIs in the three decision-making structure conditions that included AI agents (Table 3 and Figure 1). A structure by event MAN-OVA (Table 4) indicated that solo AIs, monitored AIs, and recommending AIs do not differ in perception of moral permission, $F(2,154) = .52$, ns, or moral fault, $F(2,154) = .07$, ns. However, the different events did change perception of the AI's moral

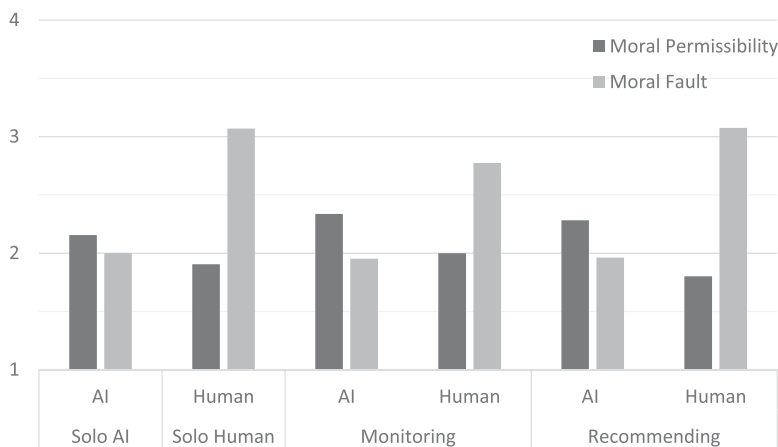


Figure 1. Mean moral permissibility and fault by structure and agent for the violation conditions only.

Table 4. F-statistics from a MANOVA predicting moral attributions.

	Human versus AIs (Solo Structures Only) (N = 116)		Artificial Intelligence (N = 168)		Humans (N = 168)		Human versus AIs (Joint Structures Only) (N = 216)	
	Permissibility	Fault	Permissibility	Fault	Permissibility	Fault	Permissibility	Fault
Structure			.52	.07	.61	3.49*	.86	2.90 [†]
Agent	1.79	63.82***					10.21**	110.23***
Event	1.36	1.39	3.65*	6.79***	2.49 [†]	1.97	6.26***	6.19***
Structure × Agent							.36	2.43
Event × Structure			1.35	1.44	2.07 [†]	3.05**	2.21 [†]	.66
Event × Agent	.07	6.21***					1.84	1.08
Structure × Agent × Event							1.57	2.11 [†]

Note: Violation conditions only (N = 221). *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, [†] $p \leq .1$.

permissibility, $F(3,154) = 3.65$, $p = .014$, and moral fault, $F(3,154) = 6.79$, $p \leq .001$. Post hoc pairwise comparisons between events indicate the AI’s role in producing the violation is more permissible in the beauty contest event (2.67) compared to the other events (parole sentencing: 2.24, marginal difference at $p = .061$; children’s videos: 1.99, $p = .002$; Twitter: 2.14, $p = .016$). The AI is less at fault for the violations in both the beauty contest (1.83) and Twitter events (1.67) compared to the parole sentencing (2.10; vs. beauty contest: marginal difference at $p = .09$; vs. Twitter: $p = .006$) and the children’s videos (2.30; vs. beauty contest: $p = .003$; vs. Twitter: $p < .001$).

How do people attribute moral violations to humans in different decision-making structures?

Parallel to previous analyses, we considered the attributions made to humans in the three decision-making structures that included humans (Table 3 and Figure 1). A structure by event MANOVA (Table 4) indicates no significant difference in moral permissibility for structure, $F(2,154) = .61$, ns, and a marginally significant difference among events, $F(3,154) = 2.49$, $p = .062$. A post hoc pairwise comparison on this marginal effect, shows

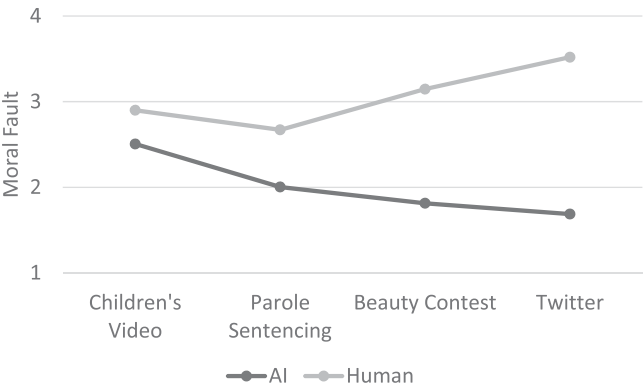


Figure 2. Mean moral fault by event and agent for the violation conditions only.

that it is more permissible for humans to produce a violation in parole sentencing (2.15) compared to the children's video event (1.79; $p = .017$). Humans' structural role in the violation change the attribution of moral fault, $F(2,154) = 3.49$, $p = .033$. Post hoc pairwise comparisons revealed that solo humans (3.07) and humans receiving recommendations from AIs (3.08) were more morally culpable than humans monitoring AI decisions (2.77; both $ps = .021$). An interaction effect qualifies this finding, $F(6,154) = 3.05$, $p = .008$, indicating this difference is most pronounced in the beauty contest and Twitter events (monitoring condition is .58 and .61 lower than the others, respectively) and near neutral in the parole sentencing and children's videos (monitoring condition is .07 lower and .10 higher than the others, respectively).

How do people attribute moral violations to AIs versus humans in different decision-making structures?

To separate and directly compare the agents from the roles they play in joint decision-making structures, we compare moral attributions towards both AIs and humans in both the recommendation and monitoring conditions (Table 3 and Figure 1) with an agent by structure by event MANOVA (Table 4). The primary finding is that both agents and events are significant for both moral permissibility and fault (agent's permissibility: $F(1,200) = 10.21$, $p = .002$; agent's fault: $F(1,200) = 110.21$, $p \leq .001$; event's permissibility: $F(3,200) = 6.26$, $p \leq .001$; event's fault: $F(3,200) = 6.19$, $p \leq .001$). When AIs and humans jointly produce a moral violation, this is more permissible for an artificial intelligence (2.31) compared to a human (1.90) and the AI is less at fault (1.96) compared to the human (2.92). Additionally, it is more permissible for the agents producing the parole violation (2.38) and the beauty contest violation (2.39) compared to the children's videos violation (1.81) and the Twitter violation (1.86; post hoc pairwise comparison show the both of the former are each higher than each of the latter at $p \leq .03$). However, the agents are more at fault for producing the parole sentencing violation (2.58) and children's video violation (2.69) than the beauty contest violation (2.23) and the Twitter violation (2.27; post hoc pairwise comparison show the former two are each higher than each of the latter two at $p \leq .017$). These findings of different AI versus human agent attributions did not deviate by event, *permissibility*: $F(3,200) = 1.84$, ns; *fault*: $F(3,200) = 1.08$, ns.

Monitoring versus recommending structures do not affect moral permissibility, $F(1,200) = .86$, ns. However, agents producing a violation in the recommending condition were marginally more at fault (2.52) than those in the monitoring condition (2.36; $F(1,200) = 2.90$, $p = .09$). There were no significant interactions between agent and decision-making structure, *permissibility*: $F(1,200) = .36$, ns; *fault*: $F(1,200) = 2.43$, ns.

Discussion

We used real-world events as the stimuli for an experiment investigating humans, artificial intelligence, and joint decision-making teams committing moral transgressions. Most notably we found that AIs across their roles in these structures do not differ in moral attributions, whereas humans in structures where they are less proximate to the decision in an oversight role are faulted less for the same moral violations. This in itself is an important finding in that the moral perceptions of advanced technologies may have more to do with taking moral fault away from humans than they do with the blame of the machines

themselves. This means that AIs on human-AI teams may have the potential to be used as moral scapegoats for human failings (Merritt et al., 2011). If decision-making arrangements allow, but do not require, humans to monitor machines – for example with self-driving vehicles – it is likely the general population would deny the human is at fault for an accident.

Furthermore, AIs are always perceived to be less at fault than the humans, mirroring other research findings (e.g., Malle et al., *forthcoming*). Even when a human was monitoring an AI the human received more blame than the AI, yet there was still some moral fault attributed to the AI being monitored. While previous research has shown that individual AIs can be attributed blame (Malle et al., 2015; Shank & DeSanti, 2018; Voiklis et al., 2016), this data indicates that they can share the blame as well. These conclusions suggest strange ideas of the downstream consequences of holding an AI teammate responsible for its actions. Will humans involved in crimes with AI ‘accomplices’ receive different sentences if blame is truly shared in the mind of the judge or jury? Could coercing an AI into a criminal or immoral act be considered similar to coercing a mentally deficient person to do the same? Human perception and attribution to AIs has profound impacts on the legal systems and implication for those technologies (Jaeger & Levin, 2016). Finally, would people then punish, rehabilitate, terminate, or take some action to return an AI teammate to proper and acceptable functioning? If AIs are unable to assimilate the blame or consequences into their internal belief system, this would appear to be a technically futile exercise simply for show.

Theoretically this study primarily builds on and contributes to the longstanding research tradition of attribution theory (Malle et al., 2014; Shaver, 1985; Weiner, 1985, 1986), but with technological actors which are not routinely considered in attribution theory. Additionally a growing body of research on morality of cryptominds – entities which are perceived to not have the same type of mind as a fully functioning human adult – is converging on ideas that attributions of moral blame are tied to perceptions of mental abilities (Gray & Wegner, 2012; Gray, Young, & Waytz, 2012; Rai & Diermeier, 2015; Waytz, Heafner, & Epley, 2014). While our study does not consider mind perception *per se*, directly comparing moral attributions to the prototypically minded humans with the cryptomind AIs adds to the evidence in this emerging research area.

Limitations and future research

Mirroring actual occurrences that people have encountered has the advantage of not being artificial, but also leaves room for uncontrolled factors within the scenarios leading to different levels of moral fault among the scenarios. Unexpectedly, we found an interesting pattern across events when comparing AI solo and human solo decision-making (Figure 2). While humans are always attributed greater fault than AIs, in situations where a human is considered to be greatly at fault, an AI is deemed to be relatively fault-free. In situations where the human is attributed less fault, an AI committing the same act is attributed more fault than it was in the situations where the human was greatly at fault. This finding suggests that the types of moral violations deemed most severely wrong for humans are the least wrong for an AI to commit. This dovetails with other research that finds different moral prescriptions for AIs compared to humans (Malle et al., 2015; Malle et al., *forthcoming*; Voiklis et al., 2016). Future research should look at a range of situations and multiple types of moral transgressions to investigate the validity of this hypothesis and the

conditions and mechanisms that cause it. Another fruitful avenue would be to consider the type of AI and the information known about its algorithms, as algorithms differ in purpose, bias, ability, and complexity (Beer, 2017; Kitchin, 2017).

This study was limited by only including two of the most common AI-human decision-making relationships. These can hardly be considered teams in the fullest sense of teamwork, but still model collaborative structures routinely found in our world. Future research should broaden its scope to include more complicated decision-making and team structures to see how moral attributions function within those. AIs manage human inputs and workloads, sometimes for large numbers of humans, in company workflows, in navigation routing, and in financial trading. Humans-in-the-loop computational models also explicitly take human input as part of their system. For recommendations, humans sometimes deal with multiple AIs competing in their recommendations or decision-making such as various smart home products trying to optimize different systems, bots in bidding wars online, or viruses and antivirus software each trying to keep control.

A concluding note about AI-Human teams

Artificial intelligence systems have influence in many domains in our society, from informing, communicating, decision-making, processing, and learning from human behavior either explicitly or implicitly. Their moral impact is underwritten by the positions they occupy, the actions they take, and humans' perceptions of and response to those actions. As AI capabilities increase these interactions transition from humans using intelligent tools to humans interacting with artificial teammates. Therefore, understanding the behavioral, moral, and perceptual implications of artificial intelligence is not simply key for understanding technology, but for more generally understanding humanity and society.

Disclosure statement

No potential conflict of interest was reported by the authors.

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